

HUDSON BAY, SK, Canada

WMO: 718680

Lat: 52.814N

Lon: 102.309W

Elev: 1175

StdP: 14.08

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.1	-24.1	-36.8	0.7	-28.7	-31.9	0.9	-23.5	21.6	9.9	19.0	10.1	4.6	210	0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	83.5	66.3	80.5	64.7	77.6	63.2	69.4	79.8	67.1	76.7	65.2	74.4	8.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	99.5	74.4	63.6	91.9	71.6	61.6	85.6	69.7	34.3	80.3	32.4	76.7	30.8	74.5	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.0	17.5	15.3	DB	-37.1	89.9	4.4	3.2	-40.3	92.3	-42.8	94.2	-45.3	96.0	-48.5	98.4	
			WB	-37.3	73.3	4.3	3.3	-40.4	75.6	-42.9	77.6	-45.3	79.4	-48.5	81.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	1.7	5.7	18.7	35.9	48.7	58.5	64.0	61.5	52.0	38.2	20.4	6.2
		DBStd	24.67	15.49	14.08	14.13	11.05	8.78	6.84	5.56	6.18	7.90	8.62	12.31	14.78
		HDD50	6979	1499	1239	970	435	133	10	2	67	377	888	1359	
		HDD65	11267	1964	1659	1435	874	508	213	86	142	393	831	1338	1824
		CDD50	1303	0	0	0	11	93	266	433	359	129	12	0	0
		CDD65	115	0	0	0	0	3	19	54	34	5	0	0	0
		CDH74	1547	0	0	0	5	78	262	608	477	114	3	0	0
		CDH80	323	0	0	0	1	13	57	123	108	21	0	0	0
		WSAvg	6.9	6.8	6.9	7.3	7.7	8.0	6.8	6.4	6.1	6.7	6.9	6.8	6.4
		PrecAvg	18.9	0.8	0.6	1.1	1.1	1.7	2.8	3.2	2.3	2.2	1.2	0.9	0.9
(15)	Precipitation	PrecMax	23.8	1.8	1.3	2.2	2.9	4.2	7.0	5.9	4.8	4.1	3.3	2.4	2.0
(16)		PrecMin	11.5	0.1	0.2	0.3	0.2	0.4	0.6	0.7	0.2	0.3	0.3	0.2	0.6
(17)		PrecStd	3.3	0.4	0.3	0.6	0.8	1.1	1.6	1.4	1.3	1.2	0.8	0.5	0.4
(18)															
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	39.6	40.8	51.6	72.3	81.5	86.1	87.4	88.2	82.8	70.6	52.1	39.9	
(20)		0.4% MCWB	35.1	35.1	42.0	53.4	59.3	67.1	69.2	69.2	63.6	55.7	43.5	34.9	
(21)		2% DB	33.8	35.5	46.8	65.6	75.5	80.5	83.2	82.7	77.1	62.4	44.7	34.2	
(22)		2% MCWB	30.9	31.9	38.6	48.9	55.7	64.0	67.1	66.6	60.9	50.6	38.8	31.3	
(23)		5% DB	28.9	30.9	42.2	59.9	71.0	76.5	80.3	79.3	72.0	56.9	39.8	29.3	
(24)		5% MCWB	27.0	28.2	35.9	46.1	53.4	61.5	65.9	64.9	58.8	47.7	35.6	27.2	
(25)		10% DB	24.4	26.5	37.7	54.1	67.0	72.9	77.3	75.8	67.0	52.0	35.9	25.7	
(26)		10% MCWB	22.8	24.3	33.2	43.1	51.8	59.4	64.4	63.2	56.2	44.9	33.1	24.2	
(27)		0.4% WB	35.5	35.7	42.2	54.0	61.4	70.5	72.7	71.7	66.1	56.7	44.0	35.4	
(28)		0.4% MCDB	39.4	39.5	50.0	70.4	73.7	82.2	82.3	83.1	77.9	68.1	51.0	39.1	
(29)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	31.4	31.7	39.0	50.0	58.2	66.2	69.9	68.9	63.1	52.4	39.4	31.4	
(30)		2% MCDB	33.7	35.1	45.7	62.9	71.6	76.2	79.9	79.8	74.1	60.5	43.9	34.3	
(31)		5% WB	27.0	28.2	36.2	46.7	55.6	63.8	67.7	66.5	60.1	48.7	35.9	27.3	
(32)		5% MCDB	28.9	30.6	41.5	58.5	68.1	73.4	76.8	75.8	69.4	55.7	39.1	29.2	
(33)		10% WB	22.7	24.2	33.5	43.7	53.0	61.4	65.7	64.6	57.3	45.7	33.1	24.2	
(34)		10% MCDB	24.3	26.2	37.5	53.1	64.5	70.1	74.8	73.4	65.8	51.4	35.9	25.6	
(35)	Mean Daily Temperature Range	MDBR	17.4	20.1	21.1	22.3	24.1	21.9	22.0	22.7	22.4	17.7	14.8	15.1	
(36)		5% DB MCDBR	17.6	20.2	22.8	31.4	31.2	27.7	26.7	28.6	31.4	27.6	19.3	16.8	
(37)		5% DB MCWBR	15.3	16.8	15.8	17.9	15.8	13.5	13.1	14.2	16.9	16.5	13.8	14.5	
(38)		5% WB MCDBR	17.5	19.2	21.5	29.0	26.9	24.7	23.6	25.2	28.6	25.7	17.9	16.0	
(39)		5% WB MCWBR	15.4	15.8	15.2	17.2	14.3	13.2	13.0	13.8	16.5	15.8	13.3	13.9	
(40)	Clear-Sky Solar Irradiance	taub	0.244	0.253	0.280	0.337	0.349	0.374	0.378	0.367	0.329	0.290	0.256	0.232	
(41)		taud	2.304	2.314	2.353	2.382	2.416	2.370	2.389	2.420	2.516	2.580	2.488	2.347	
(42)		Ebn at Noon	244	278	292	285	286	278	275	272	272	264	243	230	
(43)		Edh at Noon	18	25	30	34	34	36	35	32	26	20	16	15	
(44)	All-Sky Solar Radiation	RadAvg	341	695	1183	1565	1782	1799	1872	1538	1085	620	300	238	
(45)		RadStd	25	46	78	124	159	137	105	134	108	88	40	29	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	-0.88	-1.44	N/A	N/A	N/A
(47)	Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page